605.486 – Android Development

Module 28
Speech

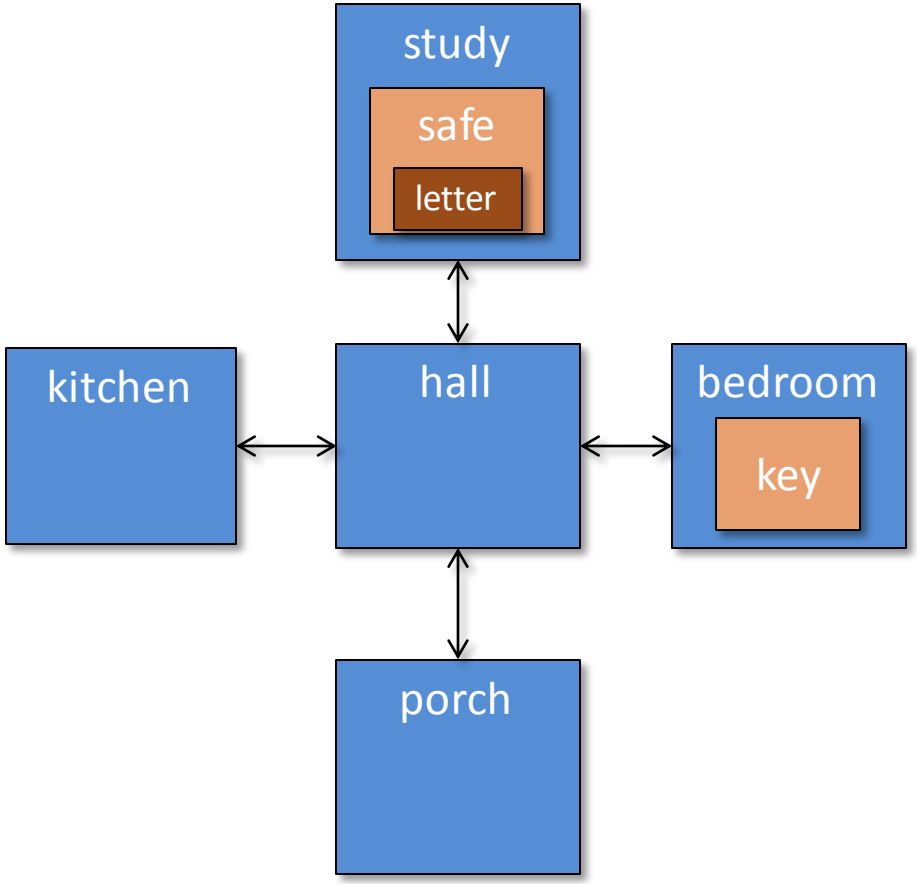
Speech

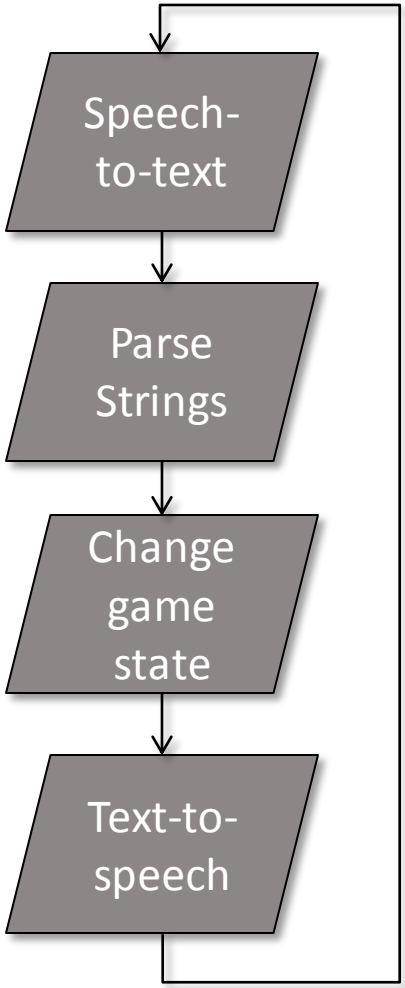
- Can generate speech from text or text from speech
- Sent normally to server to interpret
- JellyBean – can install local recognition

Speech

- Basic Speech to Text
  - Create Intent: RecognizerIntent.*ACTION_RECOGNIZE_SPEECH*
    - *Add language model (freeform vs web search) and prompt as extras*
  - startActivityForResult
  - Grab RecognizerIntent.EXTRA_RESULTS
    - Can look at EXTRA_CONFIDENCE_SCORES

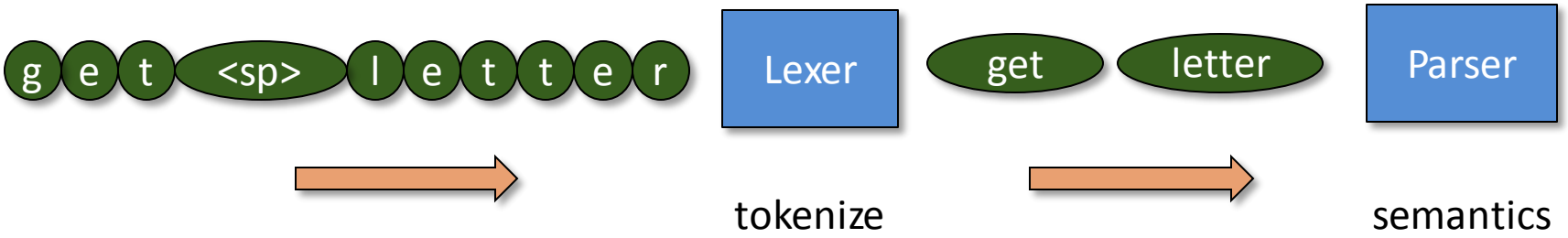
Adventure Game Example





ANTLR

- ANother Tool for Language Recognition
- Write grammar that recognizes language
- Letters -> Tokens (“words”)
- Tokens -> phrases (“sentences”)
- http://antlr.org



Adding Text to Speech

- Use TextToSpeech instance
  - speak
    - QUEUE_ADD – queue up next thing to say
    - QUEUE_FLUSH – skip rest of current and start speaking
- Initializing TextToSpeech
  - Pass OnInitListener to setup once it's ready

Storing Data in a File

- We'll use JSON for data storage
- We'll put it in the raw directory
  - No substitutions done
- Read JSON text in Game
- Read JSON file into text in Activity
- Can use JSONLint or use local validator

Automated Speech-to-Text

- Automatically listen – no button!
- BATTERY!!!
- Speech recognizer can hear sound output!
  - Need to toggle “mode” of listening/speaking

- In manifest
  - <uses-permission android:name="*android.permission.RECORD_AUDIO*"/>
  - *NO MICROPHONE PROMPT USED IN THIS EXAMPLE*

Listeners

- OnUtteranceCompletedListener UtteranceProgressListener
  - To know when text-to-speech is done to start listening
- speechRecognizer.startListening(recognizerIntent);
  - Listens for speech and triggers intent to recognize it